

Connect tables, rules, and graphs in functional thinking

Step 10

Middle School



Today I am learning to connect tables, rules, and graphs to describe and understand relationships.

1. Complete the Table Use the rule to fill in the missing outputs.

 Rule: $\text{Output} = \text{Input} + 2$

Input	1	2	3	4	5
Output	3	_____	_____	_____	_____

 Rule: $\text{Output} = \text{Input} \times 3$

Input	1	2	3	4	5
Output	3	_____	_____	_____	_____

3. Match the Table to the Rule Draw a line to match.

A

Input	1	2	3	4
Output	3	5	7	9

 • $\text{Output} = \text{Input} + 2$

B

Input	1	2	3	4
Output	5	7	9	11

 • $\text{Output} = \text{Input} + 4$

C

Input	1	2	3	4
Output	2	4	6	8

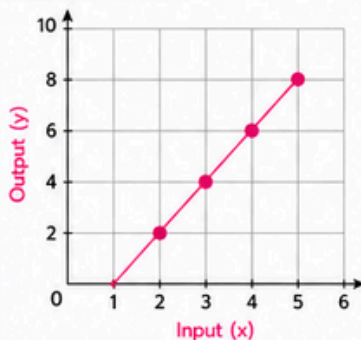
 • $\text{Output} = \text{Input} \times 2$

D

Input	1	2	3	4
Output	3	6	9	12

 • $\text{Output} = \text{Input} \times 3$

5. Read the Graph Look at the graph and answer.



- What happens when the input increases by 1?

- Is the pattern increasing or decreasing?

- What rule could match the graph?

7. Real-Life Function

A game arcade charges:

- \$5 entry fee
- \$2 per game

Complete the table.

Games	0	1	2	3	4
Cost (\$)	5	_____	_____	_____	_____

Write the rule.

Cost = _____



I can...

- ☐ connect tables, rules, and graphs.
- ☐ identify patterns and write rules.
- ☐ interpret relationships in different ways.

2. Find the Rule Look at each table. Write the rule.

Input	1	2	3	4
Output	5	6	7	8

Rule:

Output = Input + _____

Input	1	2	3	4
Output	2	4	6	8

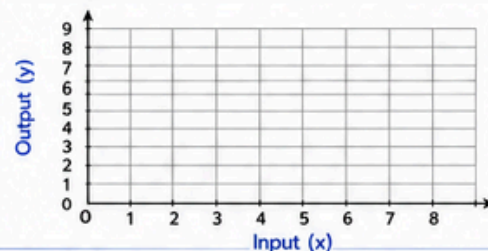
Rule:

Output = _____

4. Plot the Relationship Use the rule to complete the table. Then plot the points.

 Rule: $\text{Output} = \text{Input} + 3$

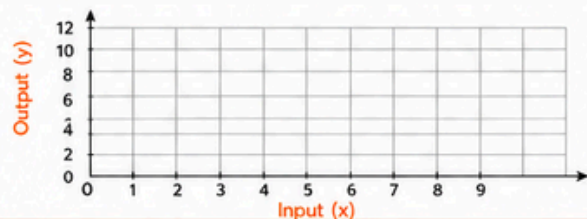
Input	1	2	3	4
Output	_____	_____	_____	_____



6. Rule → Table → Graph Use the rule to complete the table. Then plot the points.

 Rule: $\text{Output} = \text{Input} \times 2$

Input	1	2	3	4	5
Output	_____	_____	_____	_____	_____



8. Think and Explain Answer the questions in complete sentences.

- How are tables, rules, and graphs connected?

- Why are graphs useful?

- How can a rule help predict values?



My Learning Goal

I can connect tables, rules, and graphs to understand and solve problems with confidence.

